Homework 30: Linear Recurrence Relations – Solving via Companion Matrix

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BCS A, E, F, G

Introduction.

A linear recurrence relation is a formula that represents a term x_n in a sequence of numbers as a linear combination of the immediately previous terms in the sequence. That is, it is a formula for which there exist scalars $a_0, a_1, \dots, a_{k-1}$ so that

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0 x_{n-k}, \qquad \forall, \qquad n \geq k\,. \tag{1}$$

The number of terms in the linear combination is called the degree of the recurrence relation (so the recurrence relation has degree k). In order for the sequence to be completely defined, we also need some initial conditions that specify the first term(s) of the sequence. In particular, for a degree-k recurrence relation we require the first k terms to be specified.

Definition 1. ***Companion Matrix.*** The companion matrix of the linear recurrence relation

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0 x_{n-k},$$

is the matrix $k \times k$ matrix

$$\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{bmatrix}.$$

Problem 1. ***Constructing Companion Matrices.*** Construct the companion matrices of the following linear recurrence relations:

a. $x_n = x_{n-1} + x_{n-2}$.

b. $x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$.

c. $x_n = x_{n-1} + 4x_{n-2} + 2x_{n-4}$.

The following main result shows how we can represent the terms in any sequence that is defined by a linear recurrence relation via powers of that relation's companion matrix.

Theorem 1. ***Solving Linear Recurrence Relations via Matrix Multiplication.***

Let $x_0, x_1, x_2, \dots$ be a sequence satisfying a linear recurrence relation with companion matrix C. For each $n = 0, 1, 2, \dots$, define $\boldsymbol{x}_n = (x_n, x_{n+1}, \dots, x_{n+k-1}) \in \mathbb{R}^k$. Then

$$\boldsymbol{x}_n = C^n \boldsymbol{x}_0\,,$$

for all $n \geq 0$.

Problem 2. Find an explicit formula for the n −th term of the sequence defined by the recurrence relation

$$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}\,,$$

for all $n \geq 3$, if $x_0 = 3, x_1 = -2$, and $x_2 = 8$.

Hint. Try to diagonalize C to get C^n which consequently leads to finding n −th term of the sequence defined by the given recurrence relation.

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Good Luck

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